D13 The GLayout Visionaries

vIdeo Presentation Link

IEEE Solid-State Circuits Society
Technical Committee on the Open Source Ecosystem (TC-OSE)

Team D13

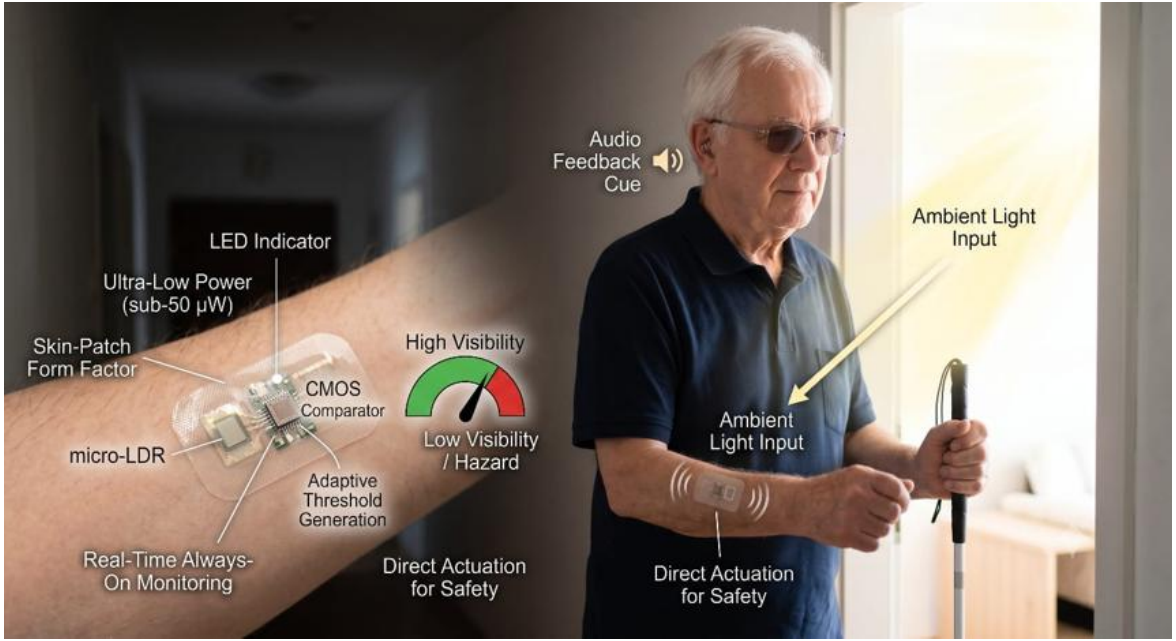
Analog Comparator for Ambient Light Detection in Wearable Assistive Devices for Visually Impaired Patients

Discord name	Affiliation	Experience	Role
iiitsavipawar_90767	Phd Student	Graduate	Team lead
bidyut02733	Btech EC 2026	Analog/digital circuit fundamentals;	Member

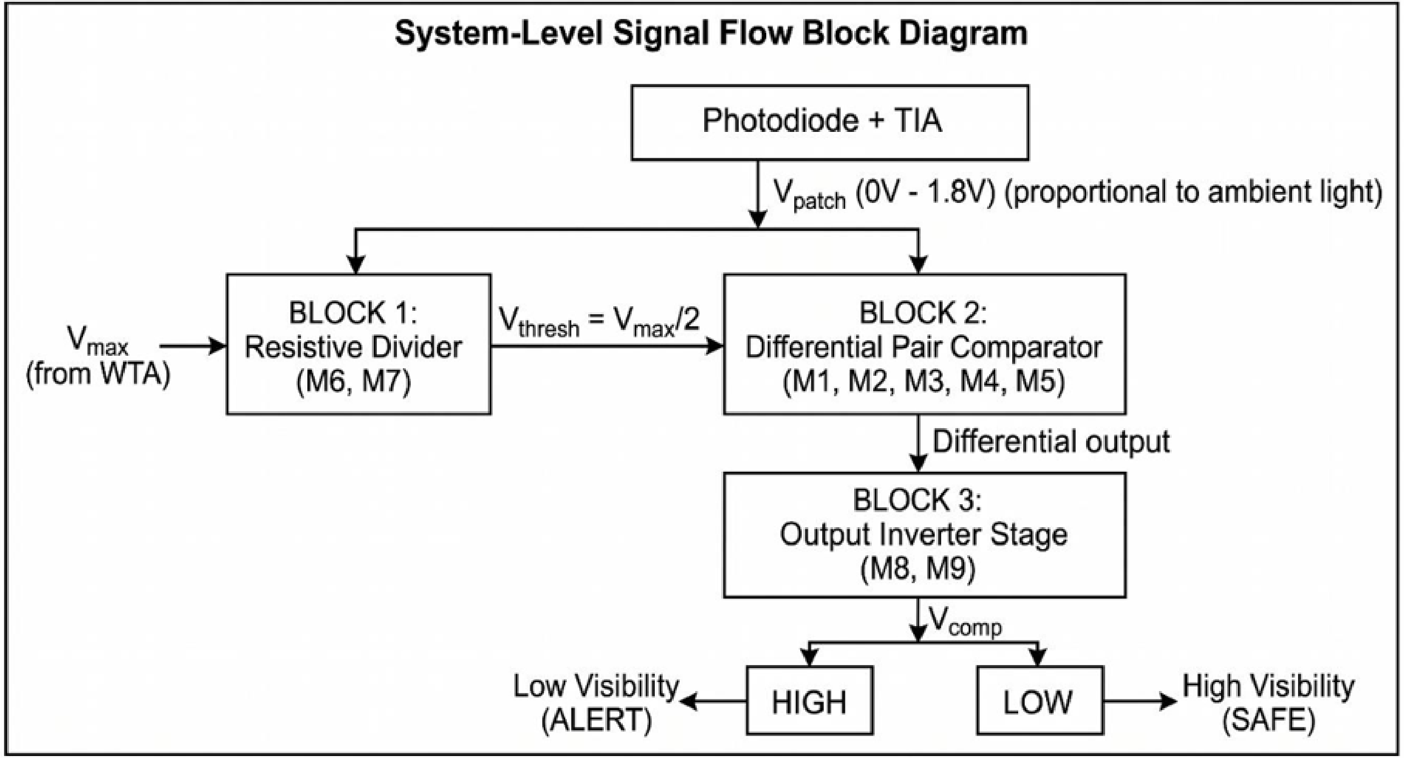
IEEE

Project Information

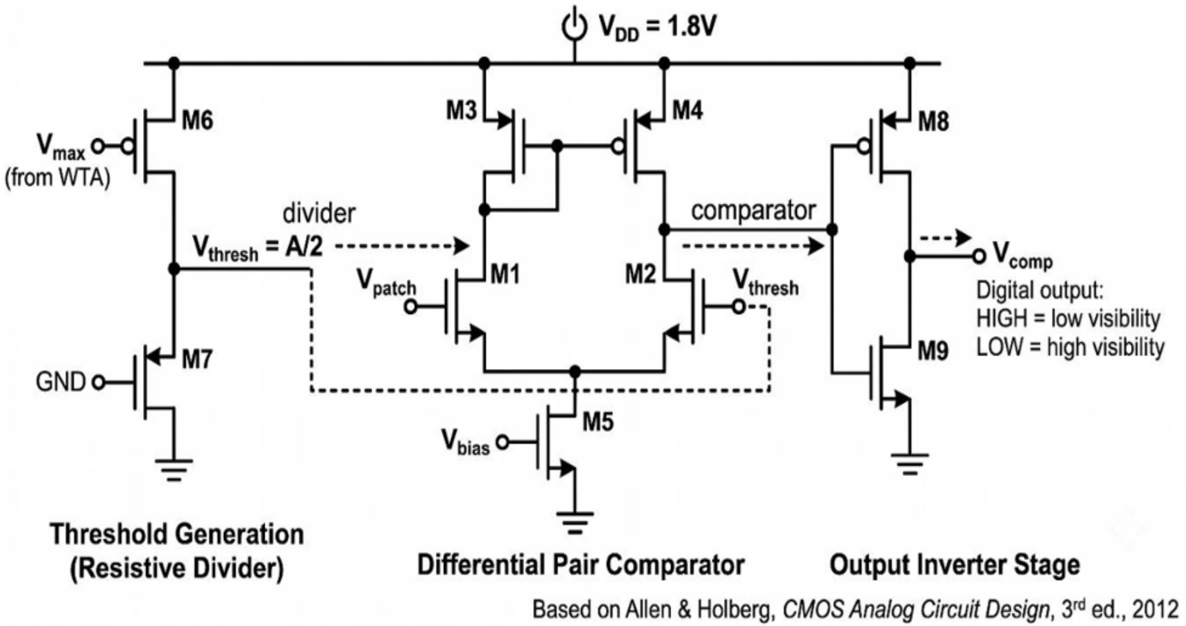
Goal: Design a fully integrated CMOS 180 nm analog comparator operating at 1.8 V for wearable skin-patch deployment. The circuit provides real-time binary ambient light classification (HIGH/LOW visibility) with sub-50 µW power consumption and adaptive threshold generation via a resistive divider.

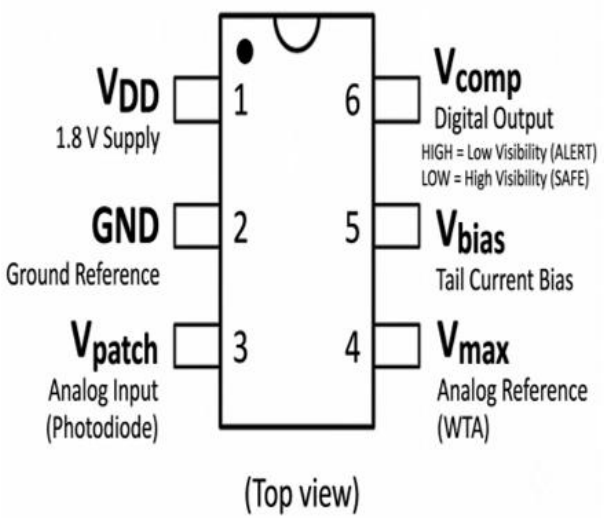


Design – High Level Proposal



Schematic Diagram



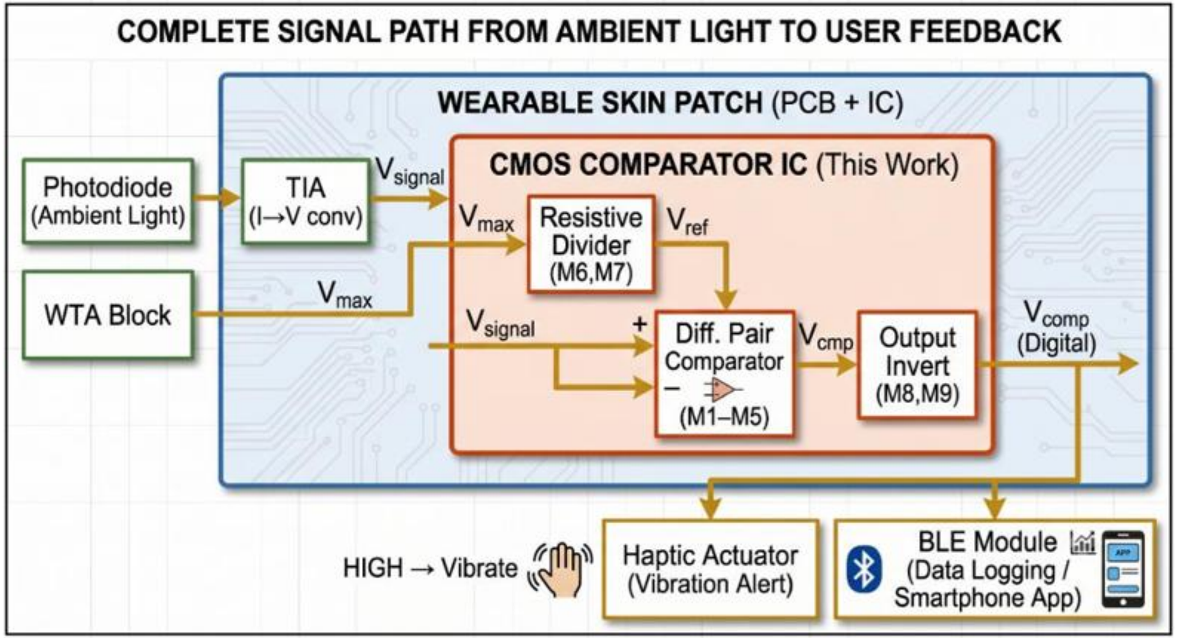


Application

Application:

Wearable assistive device for visually impaired patients — skin-patch light sensor providing haptic/audio alerts for ambient light transitions

(dark corridor, office, outdoor) targeting One year + days battery life on CR2032 at <50 µW total power.



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Team background

- Academic Experience:

  Avinash Pawar – **PhD Scholar at IIIT Surat;**

- Visiting Researcher, IIT Roorkee (NPTEL Pre-Doc Fellowship 2026);
- IEEE SSCS Member; 37+ GitHub repos;
- Research focus: near-sensor analog processing, AI/LLM-driven circuit optimization (GLayout Framework, SkyWater sky130 / GF180 PDK).

  Bidyut Maishal – BTech EC 2026,

- ALIAH University;
- Analog/digital circuit fundamentals;
- open-source EDA contributor

References

1. WHO World Report on Vision, 2019.
2. Stanchieri et al., 1.8V TIA for CMOS optoelectronics, Electronics 2022.
3. Allen & Holberg, CMOS Analog Circuit Design, 3rd ed.
4. Stanchieri et al., 180nm CMOS sensing system, Electronics 2022. Full list: github.com/sscs-ose/sscs-chipathon-2026/issues/124

Questions,
Suggestions,
Doubts?

Thank You